

This assignment will not be graded and is only for practice.

Level 1

1) Which of the following vectors can be written as the linear combination of vectors **1 point**
 $(2, 3)$ and $(1, 2)$ in $\mathbb{R}^2$ with usual addition and scalar multiplication?

- ☐ $(1, 0)$
- ☐ $(0, 1)$
- ☐ $(0, 0)$
- ☐ Any vector in $\mathbb{R}^2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 0)$
 $(0, 1)$
 $(0, 0)$
Any vector in $\mathbb{R}^2$

2) Suppose the set $\{(3, 7, 4), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ **1 point**
with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v ?

- ☐ $(0, 0, 0)$
- ☐ $(5, \frac{35}{3}, \frac{20}{3})$
- ☐ $(1, 0, 0)$
- ☐ $(6, 14, -8)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, 0)$
 $(5, \frac{35}{3}, \frac{20}{3})$

3) Suppose the set $\{(2, 3, 0), (0, 1, 0), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v ? **1 point**

- ☐ $(2, 3, 1)$
- ☐ $(1, 2, 0)$
- ☐ $(1, 3, 0)$
- ☐ $(0, 1, 1)$
- ☐ $(\pi, e, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2, 0)$
 $(1, 3, 0)$
 $(\pi, e, 0)$

4) If $(3, 4) = \alpha(1, 2) + \beta(1, 3)$ in the vector space $\mathbb{R}^2$ with usual addition and scalar multiplication, then find the value of $\alpha + 3\beta$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

5) If $(9, 3, 1)$ is a linear combination of the vectors $(1, 1, 1)$, $(1, -1, 1)$ and (x, y, z) as $2(1, 1, 1) + 3(1, -1, 1) + 4(x, y, z) = (9, 3, 1)$, then find the value of $x + y + 2z$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

6) Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ i.e. $v_1 = av_2 + bv_3 + cv_4$ then **1 point**

- ☐ v_2 is a linear combination of $v_i, i = 1, 3, 4$.
- ☐ v_3 is a linear combination of $v_i, i = 1, 4$.
- ☐ v_2 is a linear combination of v_2 .
- ☐ v_4 is a linear combination of v_4 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

v_2 is a linear combination of v_2 .
 v_4 is a linear combination of v_4 .

7) Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$ in the vector space $\mathbb{R}^3$, with usual addition and scalar multiplication. Which of the following sets is (are) true? **1 point**

- ☐ $\{v_1, v_2, v_3\}$ is a linearly dependent set.
- ☐ $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.
- ☐ $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.
- ☐ $\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.
 $\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

Level - 2

8) Let S be a subset of $\mathbb{R}^3$ which is linearly dependent. Which of the following options **1 point** are true?

- ☐ $S \cup \{(1, 0, 0)\}$ must be linearly dependent.
- ☐ $S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
- ☐ $S \setminus \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
- ☐ There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

$S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

$M_{3 \times 3}(\mathbb{R})$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices: $M_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$, $M_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $M_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$, $M_4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$

Use the above information to answer questions 9 and 10.

9) Which of the following options are correct?

1 point

- ☐ $M_1 - M_2 = 0$, where 0 denotes the zero matrix of order 3.
- ☐ $2M_1 + M_2 = M_3$
- ☐ $2M_1 - M_2 - M_3 = 0$, where 0 denotes the zero matrix of order 3.
- ☐ $2M_2 - M_1 - M_4 = 0$, where 0 denotes the zero matrix of order 3.
- ☐ The tuple (a, b, c) is unique for which $aM_1 + bM_3 + cM_4 = 0$ holds, where 0 denotes the zero matrix of order 3.
- ☐ The pair (a, b) is unique for which $aM_1 + bM_2 = 0$ holds, where 0 denotes the zero matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2M_1 - M_2 - M_3 = 0$, where 0 denotes the zero matrix of order 3.

$2M_2 - M_1 - M_4 = 0$, where 0 denotes the zero matrix of order 3.

The pair (a, b) is unique for which $aM_1 + bM_2 = 0$ holds, where 0 denotes the zero matrix of order 3.

10) Which of the following options is(are) true?

1 point

- ☐ $\{M_1, M_2\}$ is a linearly dependent set.
- ☐ $\{M_1, M_2, M_3\}$ is a linearly dependent set.
- ☐ $\{M_1, M_2, M_4\}$ is a linearly dependent set.

☐ $\{M_1, M_3, M_4\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2, M_3\}$ is a linearly dependent set.

$\{M_1, M_2, M_4\}$ is a linearly dependent set.

$\{M_1, M_3, M_4\}$ is a linearly dependent set.

Your score is: 0/10